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#### MULTI-WAREHOUSE & STORE VERIFICATION ENGINE

## Module 3: Store Verification, Stock Control & Location Management

Comprehensive 15-week implementation roadmap for Inward Store Verification (PVN), 60/40 sampling daily stock audits, 5-tier location spatial layout, multi-branch stock transfers (GIT) with handheld QR scanning, and FIFO valuation for **Popular Auto Spare & A/C Parts Co. W.L.L.**

TOTAL MODULE FEE

**BD 5,113.636**

IMPLEMENTATION TIMELINE

**15 Working Weeks**

DELIVERY EFFORT

**3.75 Engineering Months**

MILESTONE TRANCHES

**4 Verified Gates (25%)**



**5 Tiers**

Zone/Aisle/Rack/Shelf/Bin



**Android PDA**

QR Scanner Sync Engine



**7-Stage GIT**

Branch Transfer Custody



**15-Day UAT**

Acceptance SLA

## Functional Scope & Warehouse Architecture Breakdown

Module 3 transforms Popular's central warehouse and multi-branch stores into real-time, scanner-driven distribution centers with zero stock loss, automated bin navigation, and rigorous daily physical count audits.

### 5-Tier Spatial Location Matrix

Precise bin coordinate tracking (Warehouse → Zone → Aisle → Rack → Shelf → Bin) with printed QR codes for instantaneous pick-and-pack routing.

### Blind 60/40 Cycle Count Engine

Dynamic physical inventory audits (60% high-value/fast-moving + 40% random SKUs) with day-closing hard locks preventing unaccounted inventory leakages.

### Goods-in-Transit (GIT) Transfer

7-stage transfer lifecycle with driver handheld confirmation, destination warehouse receiving scans, and automatic variance logging.

### Min/Max Auto-Replenishment

Automated branch reorder alerts calculated from 90-day sales velocity, seasonal demand spikes, and supplier lead times.

## Dedicated Engineering Team & Resource Rate Matrix (BHD)

Pricing for Module 3 is calculated strictly using the dedicated 5-person engineering team across 15 working weeks (3.75 calendar months):

**Project Manager & Solutions Architect**

**BD 545.455**

BD 136.364 / Wk • 3.75 Mo = BD 2,045.456

100% Dedicated

**Back-End Lead Engineer (PHP MVC)**

**BD 227.273**

BD 56.818 / Wk • 3.75 Mo = BD 852.274

100% Dedicated

**Front-End Lead Engineer (React 18)**

**BD 227.273**

BD 56.818 / Wk • 3.75 Mo = BD 852.274

100% Dedicated

**Database & Cloud DevOps Specialist**

**BD 204.545**

BD 51.136 / Wk • 3.75 Mo = BD 767.044

100% Dedicated

**QA Automation & Test Lead**

**BD 159.091**

BD 39.773 / Wk • 3.75 Mo = BD 596.591

100% Dedicated

| ENGINEERING DESIGNATION                           | MONTHLY RATE (BHD) | WEEKLY RATE (BHD) | HOURLY RATE (160H) | EFFORT ALLOCATION    | TOTAL COST (BHD) |
|---------------------------------------------------|--------------------|-------------------|--------------------|----------------------|------------------|
| Project Manager & Enterprise Solutions Architect  | BD 545.455         | BD 136.364        | BD 3.409           | 3.75 Months (15 Wks) | BD 2,045.456     |
| Back-End Lead Engineer (PHP MVC / REST APIs)      | BD 227.273         | BD 56.818         | BD 1.420           | 3.75 Months (15 Wks) | BD 852.274       |
| Front-End Lead Engineer (Admin UI / React Tokens) | BD 227.273         | BD 56.818         | BD 1.420           | 3.75 Months (15 Wks) | BD 852.274       |
| Database & Cloud Infrastructure Architect         | BD 204.545         | BD 51.136         | BD 1.278           | 3.75 Months (15 Wks) | BD 767.044       |
| QA & System Test Automation Lead Engineer         | BD 159.091         | BD 39.773         | BD 0.994           | 3.75 Months (15 Wks) | BD 596.591       |
| TOTAL MODULE 3 DEDICATED ENGINEERING INVESTMENT   |                    |                   |                    | 15 Working Weeks     | BD 5,113.636     |

Milestone Delivery Breakdown & Payment Tranches

| Milestone Phase               | Deliverable Description & Scope                                             | Timeline    | Share (%) | Invoice Amount (BHD) | Acceptance Verification Trigger                                                |
|-------------------------------|-----------------------------------------------------------------------------|-------------|-----------|----------------------|--------------------------------------------------------------------------------|
| Milestone 1                   | Warehouse Spatial Hierarchy (5 Tiers), Bin Matrix & Inward PVN Setup        | Weeks 1-4   | 25.00%    | BD 1,278.409         | Demonstration of Warehouse spatial master, Bin QR generator & Inward PVN       |
| Milestone 2                   | Handheld Scanner Engine, Blind 60/40 Cycle Counts & Variance Alerts         | Weeks 5-8   | 25.00%    | BD 1,278.409         | Sign-off of Android PDA scanner integration & blind stock audit reconciliation |
| Milestone 3                   | Inter-Branch Stock Transfers (GIT), Multi-Tier Discrepancy & FIFO Valuation | Weeks 9-12  | 25.00%    | BD 1,278.409         | Verification of 7-stage GIT transfers, driver sign-off & FIFO valuation math   |
| Milestone 4                   | Min/Max Safety Stock Auto-Replenishment, Aging & Final Warehouse Sign-Off   | Weeks 13-15 | 25.00%    | BD 1,278.409         | Formal multi-warehouse UAT sign-off and final production deployment            |
| TOTAL MODULE 3 CONTRACT VALUE |                                                                             | 15 Weeks    | 100.00%   | BD 5,113.636         |                                                                                |

## Terms, SLA & Governance Protocols

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### 1. 15-Day UAT Review SLA

- Popular Auto Spare is allocated a 15-calendar-day UAT review window per milestone to test all warehouse scanning and inventory transfers.

### 2. Source Code & IP Transfer

- Full proprietary ownership of warehouse spatial models, barcode scanner event listeners, and audit engines transfers to Popular Auto Spare Co.

# Stakeholder Authorization & Digital Sign-Off Console

By affixing digital signatures below, authorized executive representatives formalize the approval and scope ratification for **Module 3: Store Verification & Stock Control Milestone.**

**Popular Auto Spare Co. W.L.L**

Client Executive Sign-off & Commercial Approval

No Signature on Record

PENDING SIGN-OFF

**SaNDS Lab Middle East W.L.L**

Lead Solutions Architect & Engineering Director

No Signature on Record

PENDING SIGN-OFF

**UniGlobal Consultancy**

Executive Strategic Advisor & Governance Lead

No Signature on Record

PENDING SIGN-OFF

